

MQTT based Air Quality Monitoring

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Abstract—Air quality plays very important role in safety, security, and health of the mankind. While increasing large-scale industrialization and urbanization creates huge cities, these activities have been creating various ill-effects to the environment. Likewise, one problem is deterioration of air quality in various Indian cities. The primary contributor to air pollution is the particulate matter (PM2.5), causing human health issues, like asthma and other respiratory illnesses. A study has found out that the persons inhaling particulate matter in air are at more risk of lung cancer and comparable to non-smoker susceptible to second-hand smoke. In this work, GP2Y1010AU0F, a sharp dust sensor is used for measuring smoke and dust particles in the environment. The measured data is sent using ESP8266 based NodeMCU-12E, a wireless micro-controller to the cloud server by using Message Queue Telemetry Transport protocol. The MQTT uses publish, subscriber, and broker services. Also, carbon monoxide sensor (MQ-7) is used for sensing carbon monoxide level in the air. The measured values of carbon monoxide (MQ-7) and of sharp dust sensor GP2Y1010AU0F are shown using web page and mobile application.

I. INTRODUCTION

The air quality in any mega city is of great concern as a consequence of massive urbanization and population growth, increase in industrialization, vehicular road traffic and congestion and energy use. The presence of pollutants in the air and continued increase of the same has become the major cause of lung related diseases to the urban dwellers due to continued exposure. According to World Health Organization (WHO), air pollution is one of the most serious threats [3] and about nearly 6.6 million people died in India due to air pollution. One of the major threats in mega cities is outdoor pollution. The air pollution is due to vehicular road traffic and congestion, thermal power plants, infrastructure developmental works, burning waste within mega-cities and improper agriculture practices used during the post harvest stage in the regions surrounding the urban areas. At present governmental agencies measure pollution levels using various sensor network. They measure various pollutant gases [4], [5] such as nitrogen oxides, carbon monoxide, sulfur dioxide and particulate matter. The measured data is validated by using instrument calibration and post-processing. By observing long duration time-series data regulatory decisions are made [6].

Particulate matter PM2.5 [9],[11] refers to fine particles that are 2.5 μm or diameter less than 2.5 micrometers. The sources of Particulate matter (especially PM2.5) are emitted from wood burning, power plants [18], industrial processes, diesel based vehicles and the same is due to fuel combustion.

Particulate matter is being considered in view of human health risk. According to scientific studies [3], Particulate matter (PM2.5) [12] causes various health problems such as asthma, bronchitis, and chronic respiratory symptoms like painful breathing and shortness of breath [3].

National Air Quality Monitoring Programme [4](NAMP) in India is executed by Central Pollution Control Board (CPCB) [4]. Air quality standards are the primary requirements for air pollution control and this standard adopted by the regular authority and implemented. The objective of these standards is protecting human health from adverse issues of air pollutants. The Central Pollution Control Board notified Indian Air Quality Standards, as shown in the Table I.

TABLE I
INDIAN NATIONAL AIR QUALITY STANDARDS [12],[4]

pollutant	Avg time(Hr)	Standard($\mu\text{g}/\text{m}^3$)
Nitrogen dioxide(NO_2),	24	80($\mu\text{g}/\text{m}^3$)
Particulate matter(PM2.5)	24	60($\mu\text{g}/\text{m}^3$)
Particulate matter(PM10)	24	100($\mu\text{g}/\text{m}^3$)
Sulfur dioxide(SO_2)	24	80($\mu\text{g}/\text{m}^3$)
Carbon monoxide(CO)	8	2(mg/m^3)

Currently, governmental agencies have been monitoring air quality in various cities across India [4] by using fixed stations, which are highly reliable and capable of measuring many air pollutant gases [11],[12]. However, fixed stations are expensive, large in size and require periodical maintenance, also installation cost of the equipment is high. The remedial solution is to use small, low cost, reliable, portable [15],[18],[20] stations [14], which are mounted on the various transportation vehicles [15], including motorcycles. Any subscriber can see the location specific pollution content by using a mobile application or the web, also they can see the location of that particular vehicle [1],[20]. The end user can also see the current location of the public transportation buses and required time of the bus arrival.

II. SYSTEM DESIGN

Figure 1 shows the architecture of the implementation. The pollution monitoring of the Environment is designed using the sharp dust sensor GP2Y1010AU0F which has 6 pins and provide analog output and also carbon monoxide sensor is used for measuring the value of CO in the environment using the MQ-7 gas sensor. Both the sensors (sharp dust

sensor GP2Y1010AU0F and MQ-7 gas sensor) are connected to the ESP8266 NodeMCU-12E. ESP8266 is then connected to the MQTT broker and the same is used for the display purpose. The sharp dust sensor GP2Y1010AU0F measures the analog data and that data is sent to the subscriber via MQTT broker. The first part consists of the sensors, Micro-controller and the second part consists of MQTT broker [6] and displaying the results. The ESP8266 NodeMCU-12E [7], used as a network gateway and is connected to the sharp dust sensor, GP2Y1010AU0F and MQ-7 gas sensor using wired connection.

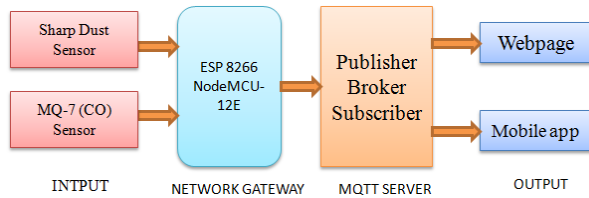


Fig. 1. SYSTEM ARCHITECTURE OF THE IMPLEMENTATION

In the second part, the user initially has to do the primary procedure of MQTT-Thingier.io server. In it the user installs Thingier.io library in Arduino-IDE by using Arduino library manager and it is tested by using simple inbuilt examples (led blinking) and it also tests that message being transmitted and received is correct between the MQTT publisher and the subscriber component [5]. So whenever the sharp dust sensor GP2Y1010AU0F and MQ-7 gas sensor measures the data, that data is sent to the Esp8266 NodeMCU-12E microcontroller. The user can now view the content to which the user has subscribed like sharp dust sensor GP2Y1010AU0F and MQ-7 gas sensor values using the above-described method. For security purpose in Thingier.io platform has inbuilt security is provided using the security protocols such as SSL and TLS. The last stage involves database management, an starting of logging methods by creating the username (unique user id) and password for accessing the required data. In short, by using this method we can strengthen the entire MQTT protocol [10] system which makes it desirable to use these methods. Then at the end the data is displayed using the mobile application and webpage.

III. PROCEDURAL STEPS

The objective is to collect the data from the sensors and send this data to the user whenever the user wants to check the required particulate matter data (sharp dust sensor GP2Y1010AU0F) and carbon monoxide level (MQ-7). The micro-controller Esp8266 NodeMCU-12E acts as a gateway server for connecting to the Internet using Wi-Fi. It acts as a network having control over the sensors which give the updates of the particulate matter value (PM 2.5), Carbon Monoxide value and the status of the components. The data is secured using security Protocol (Transport layer security). We have used a sharp dust sensor GP2Y1010AU0F(which measures analog

value) and carbon monoxide level MQ-7(which measures both analog and digital values) for our project. Moreover, the data is stored continuously in the MQTT Server. The advantage of using MQTT protocol and Secure Socket Layer (SSL) cryptographic protocol is that no ambiguous data is stored along with the required data and security is also maintained.

IV. HARDWARE DESCRIPTION

A. ESP8266 NODEMCU

The Esp8266 nodemcu [16] is one of the most popular inexpensive boards supporting wireless connectivity. Esp8266 nodemcu is built by Espressif and it contains USB connector for providing power supply. This device has a 32-bit processor, Tensilica LX 106. ESP8266 nodemcu supports Lua scripting language and Arduino IDE. Esp8266 nodemcu device is low cost and featuring extra low power consumption. The device cost is under \$3.

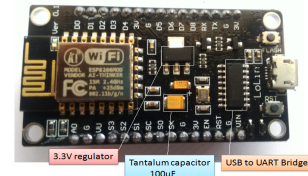


Fig. 2. ESP8266 NoDEMCU-12E

B. SHARP DUST SENSOR GP2Y1010AU0F

The sharp dust sensor GP2Y1010AU0F is a more accurate and low price device, used for measuring smoke and dust particles [17]. The sharp dust sensor GP2Y1010AU0F is an optical dust sensor, or also known as optical air Quality sensor. The sharp dust sensor consumes less power, and provides output highly reliable and stable output. The Capacitor of value 220 μ F and the Resister of value 150 Ω are used for pulse driving of given sensor led.

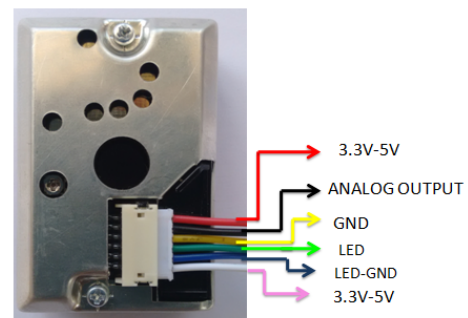


Fig. 3. PIN DEFINITIONS OF SHARP DUST SENSOR GP2Y1010AU0F

C. GAS DETECTION SENSOR - MQ-7

The MQ-7 [16] gas detection sensor is a low cost, high sensitive, reliable sensor, which is used for detecting the value of carbon monoxide gas in the environment[2],[19]. The MQ-7 gas detection sensor provides analog as well as digital output

[16], so if air is clean then it gives an output of 0.1V - 0.3V and if carbon monoxide percentage is higher then it gives around 4V analog output. This sensor can detect various other gases which contain carbon monoxide. The MQ-7 gas detection sensor requires 20s of time for warm up [16].

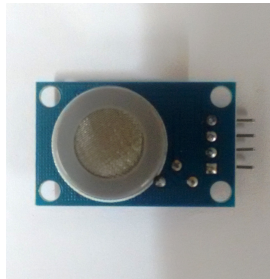


Fig. 4. GAS DETECTION SENSOR-MQ7

V. MQTT PROTOCOL

The message queue telemetry transport protocol (MQTT), used in Internet of Things (IoT) for device to device communication. Basically, MQTT contains publisher, broker, and subscriber [8] components. The MQTT protocol [6] works in Publish/Subscribe architecture. The first component in MQTT protocol [6] is the publisher, it contains sensors which are useful for measurement purpose and this measured data is sent to the broker for further processing. The second component of MQTT protocol is the broker, which is used for providing communication between publisher and subscriber. The subscriber makes use of the data.



Fig. 5. MQTT Protocol

VI. HARDWARE AND SOFTWARE ASPECTS

As shown in Figure 6, sharp dust sensor GP2Y1010AU0F and MQ-7 (carbon monoxide gas sensor) are connected to Micro-controller ESP8266 nodemcu. The analog pin (A0) of Esp8266 nodemcu is connected to the sharp dust sensor which is used for measuring particulate matter in the environment and LED pin of sharp dust sensor is connected to the GPIO13 pin (D7). The MQ-7 gas sensor pin (digital output pin- D0) is connected to the GPIO4 (D2) of the ESP8266 nodemcu.

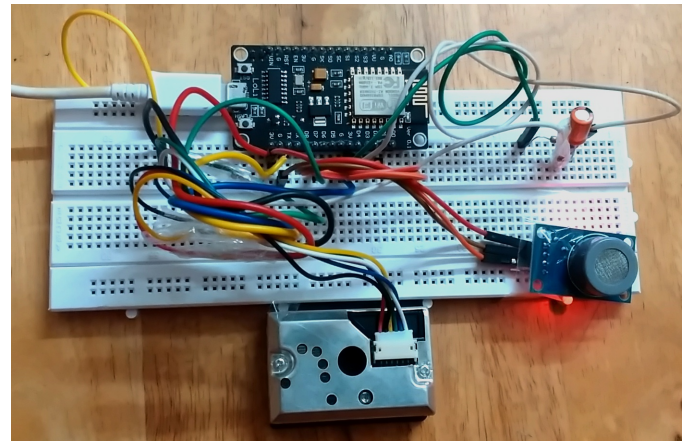


Fig. 6. Connections diagram of the system

In this proposed system Arduino-IDE is used as the same supports Thingier.io library or MyDevices.com and by using that it very simple to write code for a particular application. Thingier.io and MyDevices.com libraries contains some basic examples codes so we can test our system by using this codes. In this proposed system Thingier.io or MyDevices.com is used as a broker, one of the key components of the MQTT protocol.

VII. RESULTS

Figure 7 shows the result part of the system using a web page. It illustrates dust sensitivity value (particulate matter PM 2.5) and a digital values of carbon monoxide are displayed. The location of ESP8266 nodemcu is also displayed on the same web page. Figure 8 provides the dust sensitivity value and carbon monoxide digital value in the mobile application, so by using the web page or mobile application [14] we can measure a pollution.

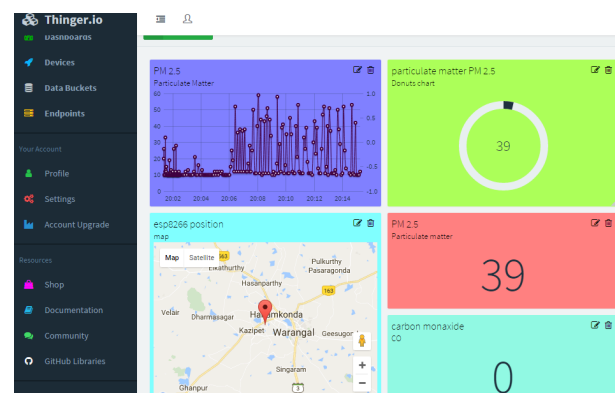


Fig. 7. Web based PM2.5 and CO status display

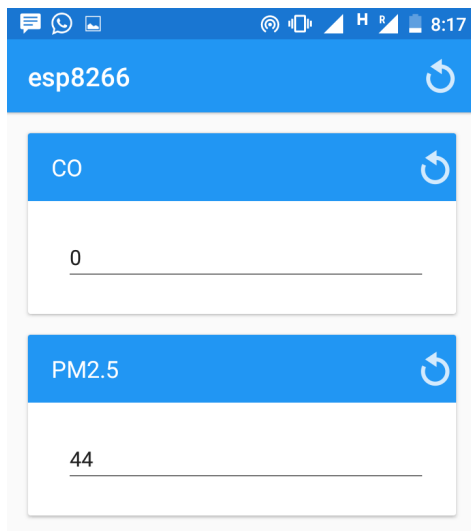


Fig. 8. PM2.5 and CO status display using mobile application

VIII. CONCLUSION

The proposed pollution monitoring system is based on WiFi micro-controller ESP8266 nodemcu, sharp dust sensor GP2Y1010AU0F (particulate matter measurement), MQ-7 (carbon monoxide measurement) sensors and MQTT protocol. By using ESP8266, sharp dust sensor GP2Y1010AU0F and MQ-7 (carbon monoxide) sensor we implemented low cost for implementation and maintenance, portable, and easy to operate. The subscriber can see the pollution content by using a mobile application. A simple pollution monitoring (particulate matter and carbon monoxide level measuring) device based on micro-controller ESP8266 nodemcu and MQTT protocol with costing under \$40 has been provided.

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